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Abstract—

I. INTRODUCTION

the detector would still output prediction, but it is not safe in those condition we didnt train for. so it is also important to know when should we stop, apart from having the high classification and localization accuracy from the detector.

we found that, The earlier stage is more sensitive to the corruption in the image than in the later stage, which is more tuned for the detector output.

different corruption have different effect on the channel reaction. For example,... (example a bit why, some are increasing the average reaction, some are making the variance bigger)

At last, we were using the first stages to calcualte the channel behavior, compare them with the closest images in the cleaning image bank (selected with the later stage, which is more about he scenario setting of the image, because it doesnt seem reasonable to compare a beach image with some busy road image for example, they definitely have very different channel reaction, and that's why we choose to use closer images.)

We validate the result for different detectors... (some numbers and trend)

II. RELATED WORK

A detector that is accurate on its test set can still fail on the road, where it meets objects and conditions it was not trained for, and it keeps producing detections all the same. Such failures, which the safety of the intended functionality (SOTIF) addresses [1], call for a runtime monitor that tells the vehicle when to stop trusting its detector [2], [3]. Monitors differ in the question they answer. Out-of-distribution (OOD) monitors ask whether the input contains an object of a class the detector was never trained on [4], [5]. Uncertainty monitors and failure predictors ask how far the detector's output can be trusted [6]–[9]. Covariate-shift monitors [10] ask whether the input itself has changed while its content stays familiar, as when the image is foggy, blurred, noisy, or compressed [11], [12]. Our work belongs to this third kind. A monitor of this kind can take its signal from one of three places: the detector's predictions (Sec. II-A), the image (Sec. II-B), or the detector's features (Sec. II-C).

A. Monitoring the Prediction

Ideally, a model should be less confident on inputs it is not familiar with [13]. The detector's outputs are therefore a natural place to start. SAOD [14] averages the uncertainty of the three most confident detections to decide whether to accept an image. ContrastiveConf [15] subtracts a weighted mean confidence of the queries a DETR [16] discards from the

mean confidence of those it keeps. Beyond the raw confidence, OOD scores computed from a classifier's logits, such as the energy [17] and the maximum logit [18], follow the same idea. These scores react only once the corruption has disturbed the detections, so a mild corruption is easily missed (Sec. IV).

B. Monitoring the Image

The image itself can be judged by a model beside the detector. Image-quality assessment (IQA) models predict how degraded an image looks. Image-based monitors differ in what they learn from: clean images, degraded images, or neither. Models that learn from clean images flag departures from them. NIQE [19] measures how far an image's local contrast statistics differ from those of clean photographs. DisCoPatch [20] trains a discriminator to reject the imperfect patches of its own variational autoencoder, and flags an image whose patches the discriminator rejects. Models that learn from degraded images either recognize particular conditions from labeled examples, such as fog [21] or rain and snow [22], or learn, from synthetically distorted images and human ratings of them, one quality score that applies to any type of degradation, as BRISQUE [23] and ARNIQA [24] do. CLIP-IQA [25] needs no quality-specific training: it rates an image zero-shot with a pretrained CLIP [26], by comparing it with the prompts "Good photo" and "Bad photo" All of these run a second model beside the detector and leave unused the features the detector has already computed.

C. Monitoring the Detector's Features

Finally, a monitor can take its signal from the detector's internal features. Hashemi et al. [27] fit a mean and spread to each neuron on clean images, and flag an image when more neurons than usual leave mean $\pm$ about two standard deviations. Becker et al. [28] compare each channel's activation distribution in a frozen detector with the channel's distribution over the training set. A learned alternative fine-tunes a separate copy of the detector's backbone, with a small embedding head, on synthetically degraded images, and flags an image whose embedding lies far from that of clean images [29].

Like Hashemi et al. [27] and Becker et al. [28], our monitor needs only the frozen detector and clean images. Their single global reference, however, mixes two sources of variation, the scene and the corruption. A mild corruption can thus hide within the differences between clean scenes. Our monitor therefore compares an image's early channels only with those of clean scenes of similar content, found by the detector's deepest stage, which reacts least to corruption yet still describes the scene (Sec. III). It also separates two effects of corruption on these channels: a flattening, where

their strongest responses fall, and a re-weighting, where their levels move up and down.

III. METHOD

IV. EXPERIMENT

V. CONCLUSION AND LIMITATION

In summary, we developed a monitor for answering the question "when is the detector not reliable anymore?".

It is usable for almost all the detector, not constraint within one type of detectors.

It has better performance (some numbers)

However, as we already discussed in sec xx, the model are trained to produce similar activation under xx from the augmentation. So even though their performance might drop, the activation cannot give us much cues, and the monitor cannot work with those yet.

REFERENCES

- [1] "Road vehicles — Safety of the intended functionality," ISO 21448:2022, Edition 1, International Organization for Standardization, Jun. 2022.
- [2] Q. M. Rahman, P. Corke, and F. Dayoub, "Run-time monitoring of machine learning for robotic perception: A survey of emerging trends," *IEEE Access*, vol. 9, pp. 20067–20075, 2021.
- [3] H. Y. Yatbaz, M. Dianati, and R. Woodman, "Introspection of DNN-based perception functions in automated driving systems: State-of-the-art and open research challenges," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 2, pp. 1112–1130, Feb. 2024.
- [4] X. Du, Z. Wang, M. Cai, and Y. Li, "VOS: Learning what you don't know by virtual outlier synthesis," in *International Conference on Learning Representations (ICLR)*, 2022.
- [5] X. Du, G. Gozum, Y. Ming, and Y. Li, "SIREN: Shaping representations for detecting out-of-distribution objects," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, 2022, pp. 20 434–20 449.
- [6] D. Feng, A. Harakeh, S. L. Waslander, and K. Dietmayer, "A review and comparative study on probabilistic object detection in autonomous driving," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 8, pp. 9961–9980, 2022.
- [7] A. Harakeh and S. L. Waslander, "Estimating and evaluating regression predictive uncertainty in deep object detectors," in *International Conference on Learning Representations (ICLR)*, 2021.
- [8] Q. M. Rahman, N. Sünderhauf, and F. Dayoub, "Per-frame mAP prediction for continuous performance monitoring of object detection during deployment," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops*, 2021, pp. 152–160.
- [9] H. Y. Yatbaz, M. Dianati, K. Koufos, and R. Woodman, "Run-time introspection of 2D object detection in automated driving systems using learning representations," *IEEE Transactions on Intelligent Vehicles*, vol. 9, no. 6, pp. 5033–5046, 2024.
- [10] J. Yang, K. Zhou, Y. Li, and Z. Liu, "Generalized out-of-distribution detection: A survey," *International Journal of Computer Vision*, vol. 132, no. 12, pp. 5635–5662, 2024.
- [11] D. Hendrycks and T. Dietterich, "Benchmarking neural network robustness to common corruptions and perturbations," in *International Conference on Learning Representations (ICLR)*, 2019.
- [12] C. Michaelis, B. Mitzkus, R. Geirhos, E. Rusak, O. Bringmann, A. S. Ecker, M. Bethge, and W. Brendel, "Benchmarking robustness in object detection: Autonomous driving when winter is coming," arXiv preprint arXiv:1907.07484, 2019.
- [13] D. Hendrycks and K. Gimpel, "A baseline for detecting misclassified and out-of-distribution examples in neural networks," in *International Conference on Learning Representations (ICLR)*, 2017.
- [14] K. Oksuz, T. Joy, and P. K. Dokania, "Towards building self-aware object detectors via reliable uncertainty quantification and calibration," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 9263–9274.
- [15] Y.-J. Park, C. Sobolewski, and N. Azizan, "Uncertainty quantification in detection transformers: Object-level calibration and image-level reliability," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pp. 1–14, 2026.
- [16] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proceedings of the European Conference on Computer Vision (ECCV)*, ser. Lecture Notes in Computer Science, vol. 12346. Springer, 2020, pp. 213–229.
- [17] W. Liu, X. Wang, J. D. Owens, and Y. Li, "Energy-based out-of-distribution detection," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 21 464–21 475.
- [18] D. Hendrycks, S. Basart, M. Mazeika, A. Zou, J. Kwon, M. Mostajabi, J. Steinhardt, and D. Song, "Scaling out-of-distribution detection for real-world settings," in *Proceedings of the 39th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 162, 2022, pp. 8759–8773.
- [19] A. Mittal, R. Soundararajan, and A. C. Bovik, "Making a "completely blind" image quality analyzer," *IEEE Signal Processing Letters*, vol. 20, no. 3, pp. 209–212, 2013.
- [20] F. Caetano, C. Viviers, L. A. Zavala-Mondragón, P. H. N. De With, and F. van der Sommen, "DisCoPatch: Taming adversarially-driven batch statistics for improved out-of-distribution detection," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, pp. 2898–2908.
- [21] M. Pavlič, H. Belzner, G. Rigoll, and S. Ilić, "Image based fog detection in vehicles," in *Proceedings of the IEEE Intelligent Vehicles Symposium (IV)*, 2012, pp. 1132–1137.
- [22] M. M. Dhananjaya, V. R. Kumar, and S. Yogamani, "Weather and light level classification for autonomous driving: Dataset, baseline and active learning," in *Proceedings of the IEEE International Intelligent Transportation Systems Conference (ITSC)*, 2021, pp. 2816–2821.
- [23] A. Mittal, A. K. Moorthy, and A. C. Bovik, "No-reference image quality assessment in the spatial domain," *IEEE Transactions on Image Processing*, vol. 21, no. 12, pp. 4695–4708, 2012.
- [24] L. Agnolucci, L. Galteri, M. Bertini, and A. Del Bimbo, "ARNIQA: Learning distortion manifold for image quality assessment," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024, pp. 189–198.
- [25] J. Wang, K. C. K. Chan, and C. C. Loy, "Exploring CLIP for assessing the look and feel of images," in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, vol. 37, no. 2, 2023, pp. 2555–2563.
- [26] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proceedings of the 38th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 139, 2021, pp. 8748–8763.
- [27] V. Hashemi, J. Křetínský, S. Rieder, and J. Schmidt, "Runtime monitoring for out-of-distribution detection in object detection neural networks," in *Proceedings of the 25th International Symposium on Formal Methods (FM)*, ser. Lecture Notes in Computer Science, vol. 14000. Cham: Springer, 2023, pp. 622–634.
- [28] S. Becker, J. Bayer, W. Hübner, and M. Arens, "Operational readiness for object detection," in *Proceedings of the 28th International Conference on Pattern Recognition (ICPR)*, ser. Lecture Notes in Computer Science, vol. 16814. Cham: Springer, 2027, pp. 121–135.
- [29] S. Becker, S. Weiss, W. Hübner, and M. Arens, "Self-aware object detection via degradation manifolds," arXiv preprint arXiv:2602.18394, 2026.